Market Basket Analysis for Understanding Customer Purchase Behavior

A Final report for the BDM capstone Project

Submitted by

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1. Executive Summary and Title

Nana Grocery Store is a neighbourhood retail outlet in Balasore, Odisha, catering primarily to daily household needs of local families. Despite steady customer footfall, the store lacked a structured understanding of purchase behaviour, category contribution, and co-purchase patterns. This resulted in suboptimal inventory planning, unclear stocking decisions, and missed opportunities for targeted promotions. The project aimed to bridge this gap by applying data-driven analysis to three months of transactional sales records.

The dataset comprised 8,067 product-level entries across 2,482 unique transactions covering 82 products and 10 categories. Descriptive statistics showed an average bill value of ₹250–₹300 and a right-skewed product price distribution dominated by low-cost essentials. The analytical workflow combined Exploratory Data Analysis (EDA) and Market Basket Analysis (MBA) using the Apriori algorithm, supported by visualizations such as monthly and daily trends, hourly patterns, category share charts, and association networks.

Key findings revealed clear seasonal and daily demand cycles, with strong morning and evening purchase peaks. Staples and Beverages emerged as core revenue drivers, while items like Atta, tea, detergents, and nutritional drinks formed the top-selling products. MBA uncovered strong co-purchase relationships such as Eggs–Potato–Wheat Rava and Rice Bran Oil–Everest Chilli, indicating predictable household buying patterns.

These insights enabled targeted recommendations including optimized inventory planning, time-based promotions, adjacency-driven shelf layout, and category-specific bundling strategies. Early implementation led to improved stock availability during peak hours and better visibility of high-demand items, demonstrating the value of a data-driven approach for small-format retail stores.

2. Detailed Explanation of Analysis Process/Method

The analytical framework for this project was developed to systematically uncover customer purchasing behaviour at Nana Grocery Store. I followed a structured workflow beginning with data collection, then cleaning and preprocessing, followed by exploratory data analysis (EDA), market basket analysis (MBA), and finally the visualization and interpretation of results. This step-by-step approach ensured that the insights generated were reliable, reproducible, and closely aligned with the store's real operational patterns.

2.1 Data Collection

The data for this project was gathered directly from the store's digital billing system and QR-payment logs covering the period from 1 August to 31 October 2025. This timeframe provided sufficient variability, including salary-period spikes, festive shopping behaviour, and weekday–weekend fluctuations. I worked with three interconnected datasets:

- Transaction Data: Contained detailed line-items for each bill, including Transaction_ID, Date, Time, Product_Name, Category, Quantity, Unit_Price, and Total_Price. Each entry represented one product purchased within a transaction.
- Product Data: Included standardized product names, categories, and unit prices, which I used as the reference during cleaning.
- Sales Summary Data: Contained product-level aggregates and category totals that helped validate the accuracy of the cleaned data.

Together, these tables provided a complete and reliable foundation for both exploratory and association analyses.

2.2 Data Cleaning and Preprocessing

Cleaning the data was especially important because any inconsistencies or errors would directly affect the accuracy of support, confidence, lift, and time-series calculations.

Standardization and Consistency Checks: I first standardized product names by mapping variations such as "Horlicks 500g", "Horlicks – 500G", and "Horlicks 500 g" to a single consistent name. This prevented identical items from being counted as separate products. Category names also required alignment, as labels such as "Home care", "Homecare", and "Home Care" appeared in different forms.

Duplicate Removal: To avoid double-counting, I identified duplicates using a composite key consisting of Transaction_ID, Date, Time, and Product_Name. Any repeated entries with the same key were removed.

Recalculation of Line Totals: To ensure arithmetic consistency, I recalculated Total_Price for each row using:
Total_Price = Quantity × Unit_Price
If a mismatch appeared, I corrected it based on the product master data.

Outlier Detection: I used the ±3σ statistical rule to detect anomalous values in Unit_Price and Total_Price. These outliers were cross-checked manually to confirm whether they were genuine or entry mistakes.

Datetime Formatting and Feature Engineering: I reformatted the Date and Time fields into proper datetime formats to enable time-based analysis. I also created additional variables including "Month", "Hour", and "Weekday" to support deeper trend exploration.

These cleaning steps ensured that product frequencies, bill values, and temporal patterns reflected the store's real business activity. Without proper preprocessing, the association rules would have been distorted and misleading.

2.3 Exploratory Data Analysis (EDA)

After cleaning, I performed EDA to understand baseline sales behaviour.

Temporal Analysis:
I aggregated sales at multiple levels:

- Daily sales: by summing Total_Price across each day
- Monthly sales: by grouping each month into total revenue
- Hourly sales: by mapping Time to hourly buckets

These calculations revealed stable weekday spending, weekend peaks, and strong morning/evening buying patterns.

Product and Category-Level Exploration: I analyzed category-wise totals, top-selling products, and price distributions. This helped identify dominant categories such as Staples and Beverages, and showed that most items fall in the low-price bracket, confirming the store's role as a neighbourhood essentials retailer.

Purpose of EDA: EDA helped me understand which products and categories appeared frequently enough to meaningfully contribute to Market Basket Analysis. It also revealed demand cycles that later supported interpretation of association rules.

2.4 Market Basket Analysis (MBA)

MBA was the core analytical method used to identify which products customers tend to purchase together. I used the Apriori algorithm, as it aligns well with transactional data from small retailers.

Transaction Encoding: I converted transactions into a binary matrix where each row represented a transaction and each column represented a product:

- 1 if the product appeared in the transaction
- 0 otherwise

This matrix format allowed Apriori to compute frequent item combinations efficiently.

Association Rule Metrics

I used three key metrics:

- Support: frequency of itemset occurrence relative to all transactions
- Confidence: likelihood that B is purchased when A is purchased
- Lift: strength of association compared to random chance

These measures helped distinguish genuine behavioural relationships from coincidental ones.

Parameter Selection and Justification:
I selected the following thresholds after experimenting with different values:

- Minimum support = 0.02 (2%)
  Ensures that an itemset appears in at least ~50 transactions, making it operationally meaningful.
- Minimum confidence = 0.40
  Ensures useful rules where at least 40% of customers purchasing A also buy B.
- Lift > 1.2
  Ensures that the relationship is stronger than random chance.

These thresholds produced rules that were statistically robust and practically relevant for a small neighbourhood store.

Apriori was chosen because it is transparent, easy to interpret, and ideal for small and medium retail datasets. The rules generated can be directly used by store owners for shelf placement and cross-selling.

2.5 Visualization and Interpretation

I created multiple visualizations to communicate insights clearly:

- Line charts for monthly and daily sales
- Bar charts for hourly sales patterns
- Histograms for product prices and bill values
- Pie charts for category share
- Network graphs for association rules

These visuals helped translate numerical outputs into insights that store owners can easily understand and act upon.

2.6 Validation and Robustness Checks

To ensure reliability:

- I re-computed rules separately for each month to check their stability
- Very low-frequency rules were removed even if they met thresholds
- I tested different support and confidence values (sensitivity analysis)
- High-lift rules were discussed with the store owner to validate their practical relevance
- These checks ensured that the final rules were not accidental or noise-driven.

Through this structured workflow, I was able to produce a detailed and reliable analysis of purchase behaviour at Nana Grocery Store. The combination of careful cleaning, thorough exploration, and rigorous association rule mining led to insights that can meaningfully support operational decision-making for the store.

3. Results and Findings

3.1 Monthly Sales Trend

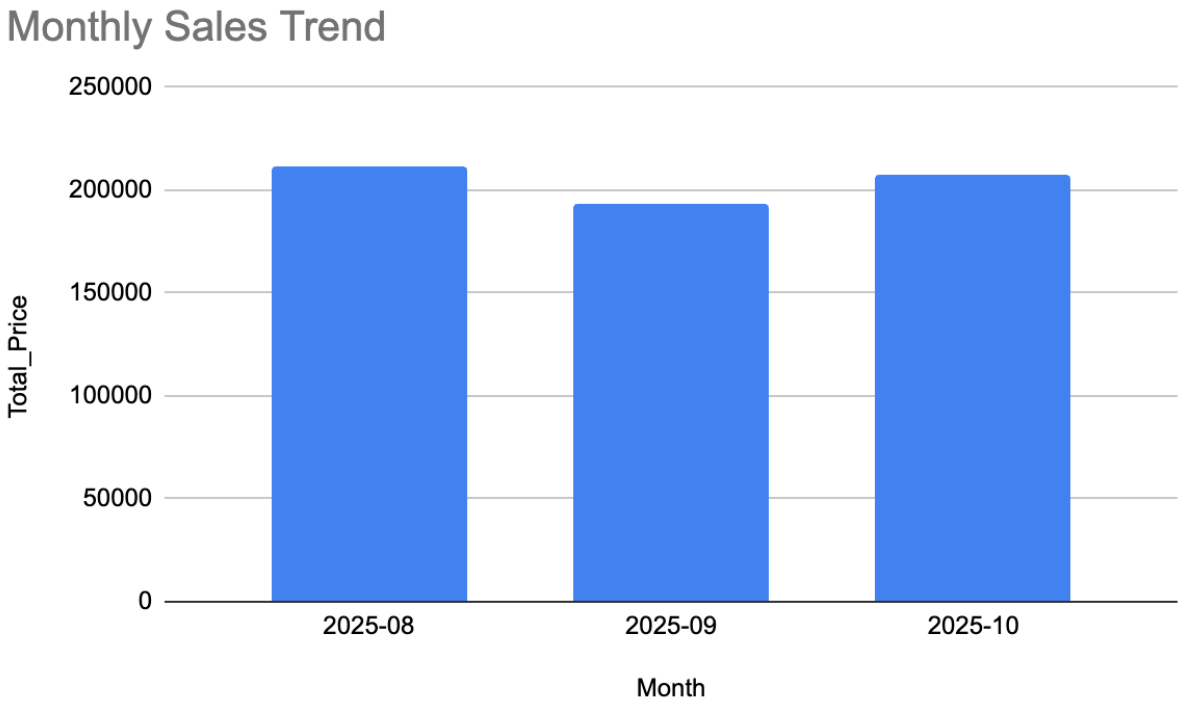


Figure 3.1: Monthly Sales Trend (August–October 2025)

- August records the highest monthly revenue (₹2,11,571), indicating strong early-quarter demand, likely driven by festival-related purchases and month-start salary cycles. This establishes August as a naturally high-performing period for the store.
- Sales dip to ₹1,93,065 in September, reflecting a mid-quarter slowdown in household spending. This decline aligns with typical post-festival normalization, suggesting that customer purchase intensity stabilizes during this period.
- October revenue rises again to ₹2,07,210, showing a clear demand recovery, which corresponds with pre-Diwali stocking and increased purchases of staples, beverages, and household essentials.
- The overall pattern shows a predictable dip–recovery cycle, where the store maintains a stable revenue baseline above ₹1.9 lakh. This consistency indicates strong recurring footfall and dependable purchasing behaviour across the quarter.
- The trend suggests actionable opportunities for the business, such as increasing inventory and promotions ahead of August and October peaks, tightening perishable stock during September, and timing combo offers to coincide with high-demand months to maximize overall revenue.

3.2 Daily Sales Trend

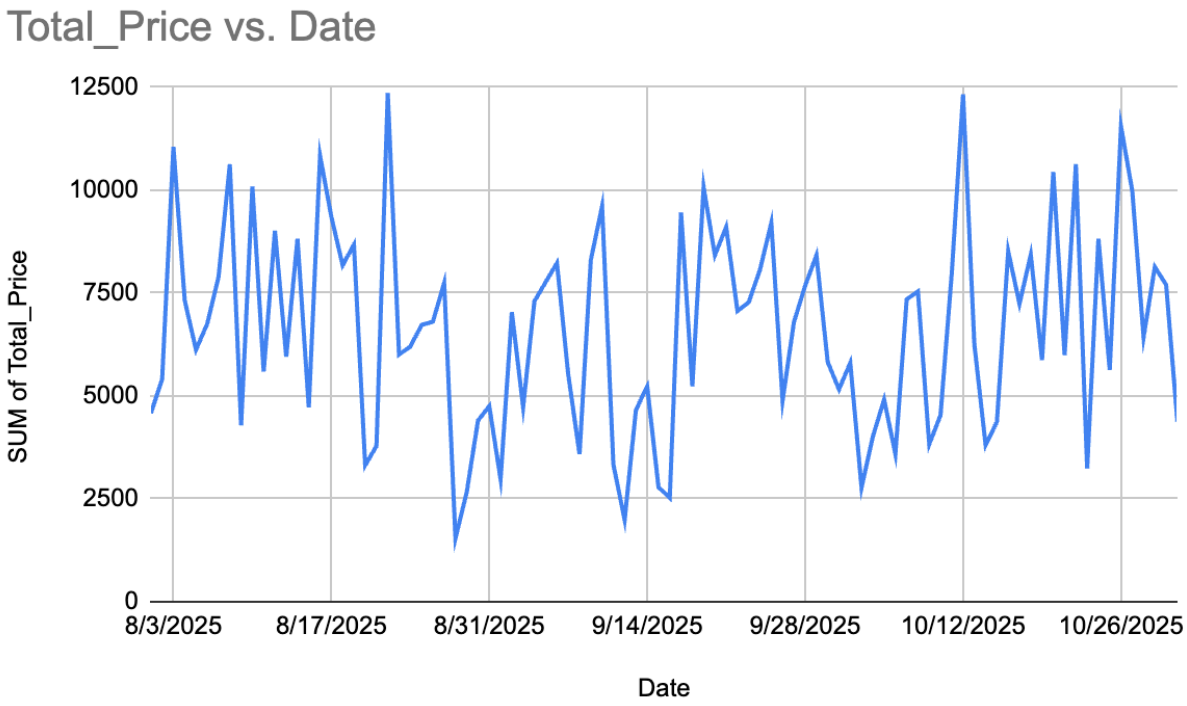
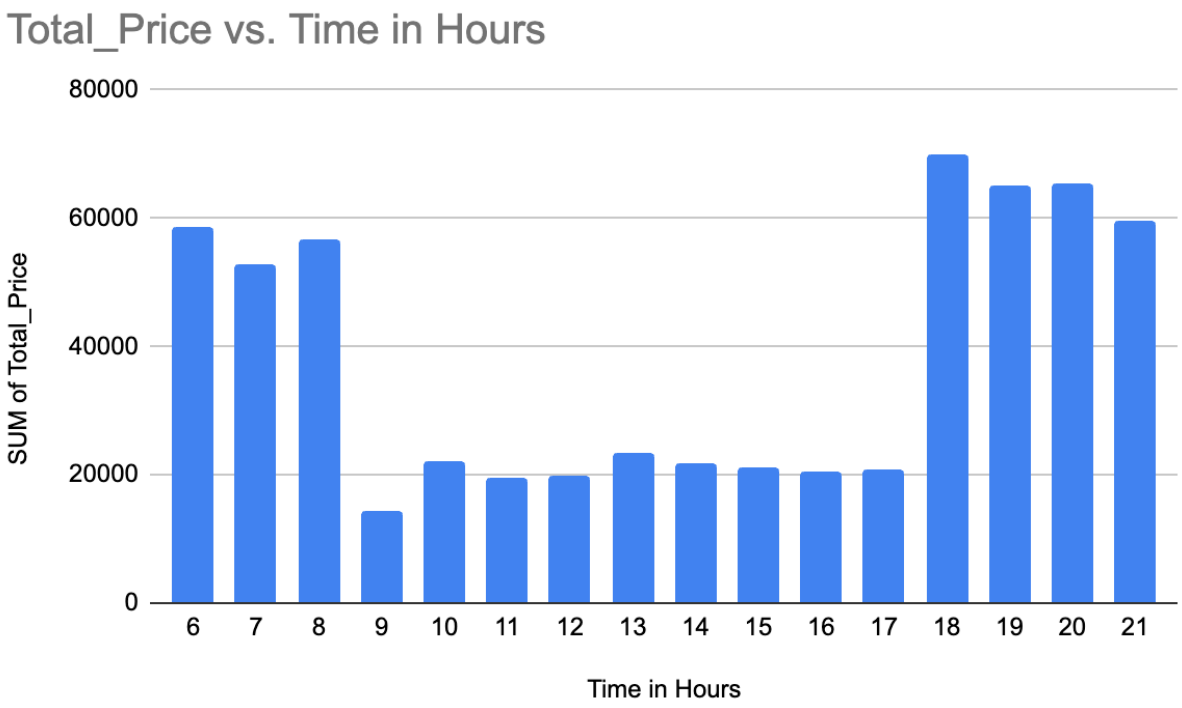


Figure 3.2: Daily Sales Trend

- Daily sales show significant fluctuations ranging between ₹2,000 and ₹12,500, indicating a highly dynamic purchasing pattern. This variation reflects the nature of neighbourhood retail, where customer flow is influenced by daily household needs rather than fixed schedules.
- Frequent spikes above ₹10,000 occur across all three months, suggesting certain days — typically weekends, festival-eve days, and salary-cycle dates — attract a higher number of purchases or higher-value baskets.
- Notable dips below ₹4,000 appear at regular intervals, which may represent weekdays with reduced footfall, customer outflow during local events, or periods of lower replenishment needs. These dips help identify slow business days that can be targeted for micro-promotions.
- The overall pattern does not show a clear linear increase or decrease, but rather a stable oscillation around the ₹6,000–₹9,000 daily band. This stability confirms consistent customer dependency on the store for essentials while also highlighting natural day-to-day demand volatility.
- The combined trend suggests clear operational opportunities, such as strengthening staffing and stocking on known high-sales days, and introducing weekday-specific offers to lift revenue on slower days. Understanding these daily cycles enables more accurate forecasting and smoother inventory turnover.

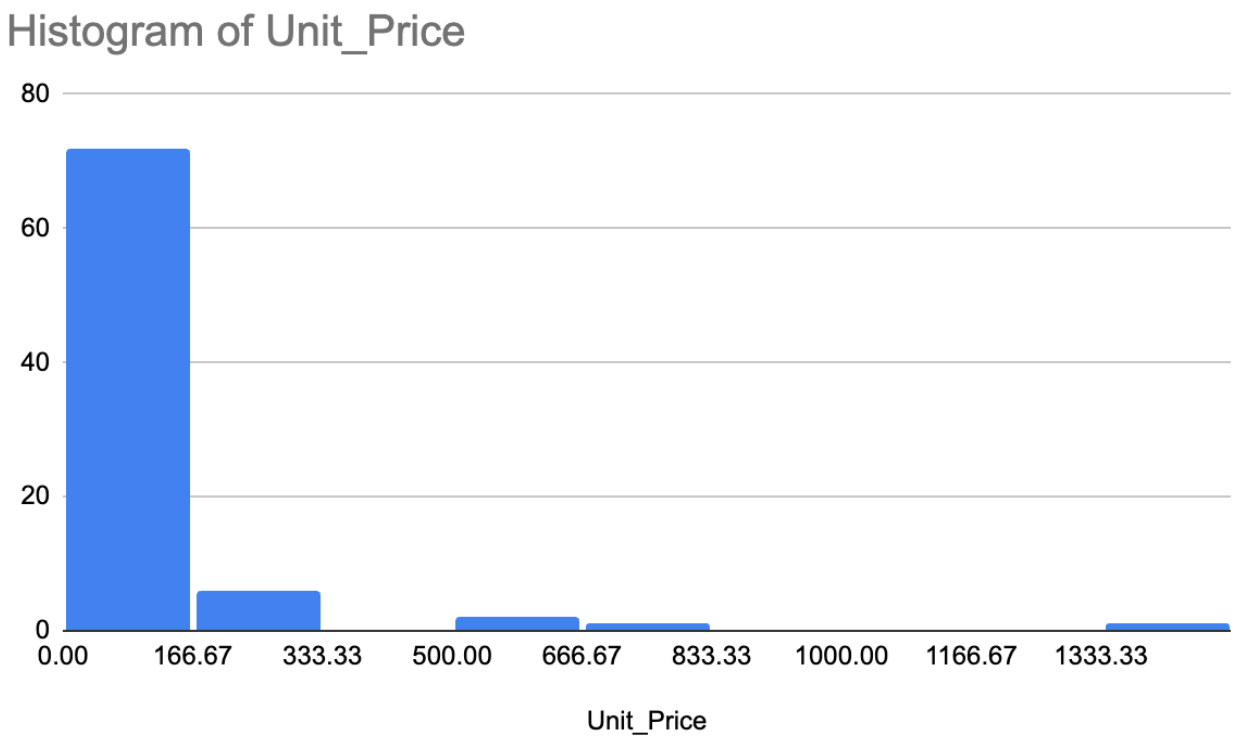
3.3 Hour-of-Day Sales Pattern

Figure 3.3: Hour-of-Day Sales Pattern



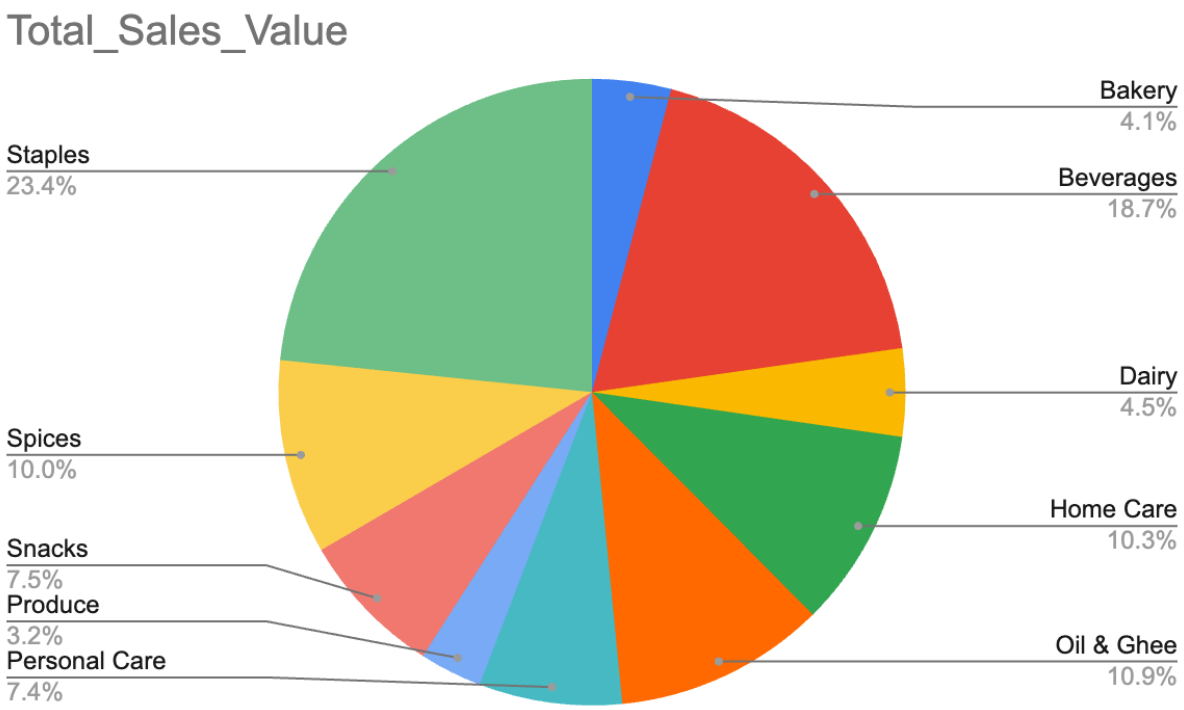
- Across the entire 3-month period, the morning hours (6 AM–8 AM) generate the highest cumulative sales, with totals between ₹52,000 and ₹60,000. This confirms that customers consistently prefer purchasing essentials early in the day before starting their routines.
- Sales volumes drop significantly during late-morning to afternoon hours (9 AM–4 PM), where cumulative sales per hour range between ₹14,000 and ₹23,000. This indicates lower footfall during working hours and a predictable midday slowdown.
- A strong second peak appears during evening hours (6 PM–9 PM), where cumulative hourly sales rise again to ₹60,000–₹70,000 across the three months. This shows that many customers prefer restocking groceries after work, creating a reliable evening surge.
- The overall pattern displays a consistent bimodal distribution across all months, with clear morning and evening peaks. This repeatable structure indicates stable daily buying habits and predictable demand cycles that can guide operational planning.
- These insights enable targeted business actions, such as increasing staff and faster checkout lanes during morning and evening peaks, scheduling shelf restocking during quieter afternoon hours, and launching time-specific promotions (e.g., evening combo discounts) to maximize revenue during the strongest demand windows.

3.4 Product Price Distribution



- The histogram shows that the majority of products fall within the lower price band of ₹0–₹150, indicating that the store's assortment is heavily skewed toward low-cost, high-rotation essentials such as staples, snacks, dairy items, and everyday household goods.
- Only a small set of products fall in the mid-range brackets (₹150–₹400), suggesting limited availability of premium or higher-priced branded goods. This reinforces the store's positioning as a neighbourhood essentials retailer rather than a premium grocery outlet.
- Very few products appear in the high-value range (₹500–₹1,300), which creates a long right tail in the distribution. These are typically bulk items, large packs, or specialized products (e.g., 5 kg atta, premium oils, nutritional supplements).
- The overall distribution is highly right-skewed, meaning low-priced items dominate the product catalogue while high-priced items contribute minimally in terms of product count. However, these higher-priced items often contribute disproportionately to revenue, which aligns with the store's sales concentration patterns found in earlier analysis.
- From a business perspective, this distribution suggests opportunities for SKU optimization—the store can strengthen mid-range and premium offerings to increase per-basket revenue, or alternatively continue doubling down on low-cost essentials by introducing combo deals and multi-buy offers tailored to budget-conscious customers.

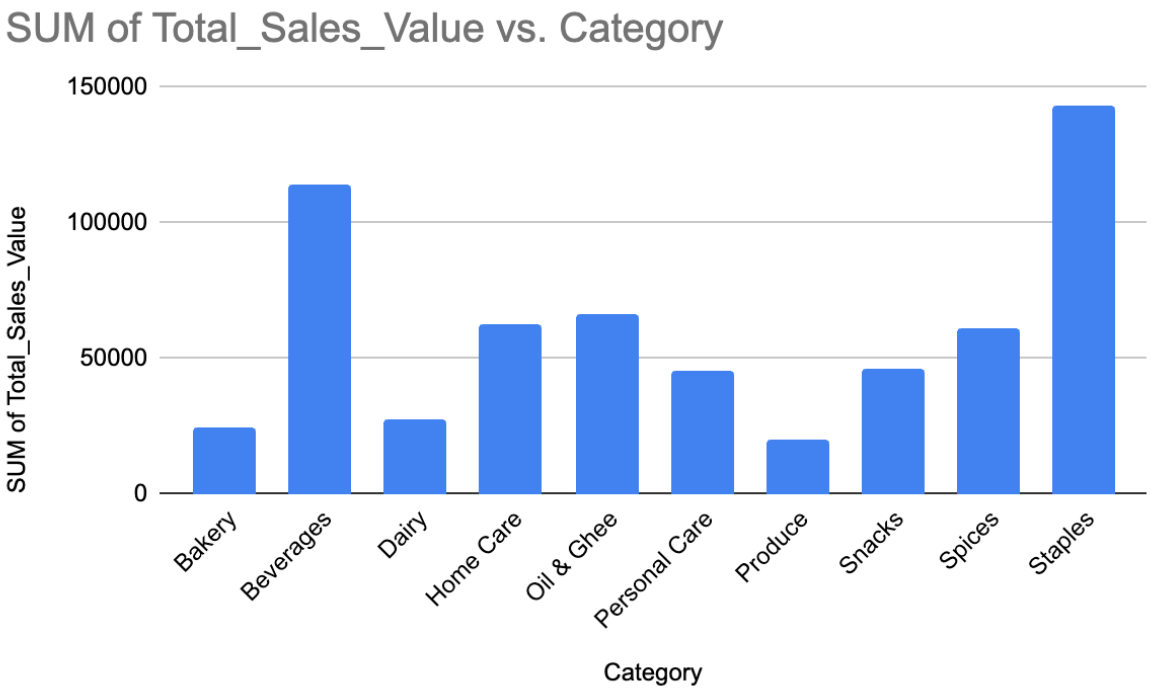
3.5 Category Share



- Staples contribute the largest share at 23.4% of total revenue, confirming their role as the primary driver of business for the store. This highlights consistent customer dependence on essential grain and household food items.
- Beverages form the second-highest share at 18.7%, reflecting high and frequent consumption of tea, coffee, and packaged drinks. The strong contribution from this category indicates a stable, repetitive buying pattern aligned with daily household routines.
- Home Care (10.3%) and Oil & Ghee (10.9%) together account for more than 21% of total sales, suggesting that non-food essentials and cooking oils have high turnover and play a significant role in regular household restocking cycles.
- Snacks (7.5%), Personal Care (7.4%), and Spices (10%) form the mid-range contribution group, indicating steady but not dominant demand. These categories offer opportunities for cross-selling with staples and beverages to increase their share.
- Low-contribution categories such as Produce (3.2%), Dairy (4.5%), and Bakery (4.1%) indicate underdeveloped sales segments, which may be due to limited assortment, perishability, or customer preference patterns. These areas provide potential for targeted

expansion through curated SKUs, improved freshness rotation, or small promotional bundles.

3.6 Category-Wise Total Sales

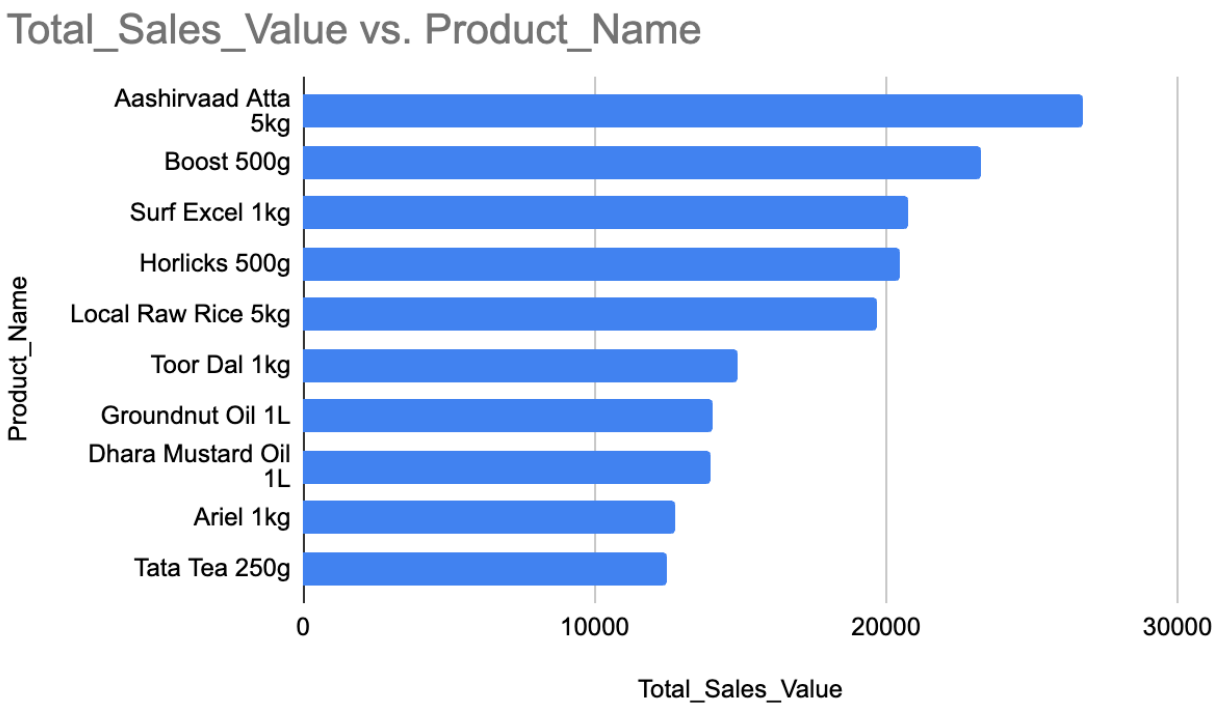


- Category Performance Overview:The bar chart displays total sales value distribution across ten product categories. Staples and Beverages emerge as the dominant revenue contributors, together accounting for the largest share of overall sales during the three-month period.
- High-Performing Categories:Staples generate the highest total sales (around ₹1.4 lakh), followed closely by Beverages (₹1.1 lakh). This pattern indicates consistent demand for essential daily-use items and high turnover of packaged drinks, aligning with regular household purchase habits.
- Moderate Contributors:Oil & Ghee, Home Care, and Personal Care categories record moderate sales levels, reflecting steady but non-seasonal buying behavior. These items are typically replenished on a biweekly or monthly basis, contributing stable revenue without significant peaks.
- Low-Contribution Categories:Produce, Bakery, and Dairy categories show comparatively lower sales volumes. This trend may stem from perishability, limited shelf life, or smaller

product range, suggesting potential areas for expansion or promotional focus to increase movement.

- Business Implications:The sales distribution reveals a dependence on core household essentials. To diversify revenue, the store can introduce cross-promotions between high-volume (Staples) and moderate-volume (Snacks or Spices) categories. This will help balance sales performance and encourage multi-category purchases.

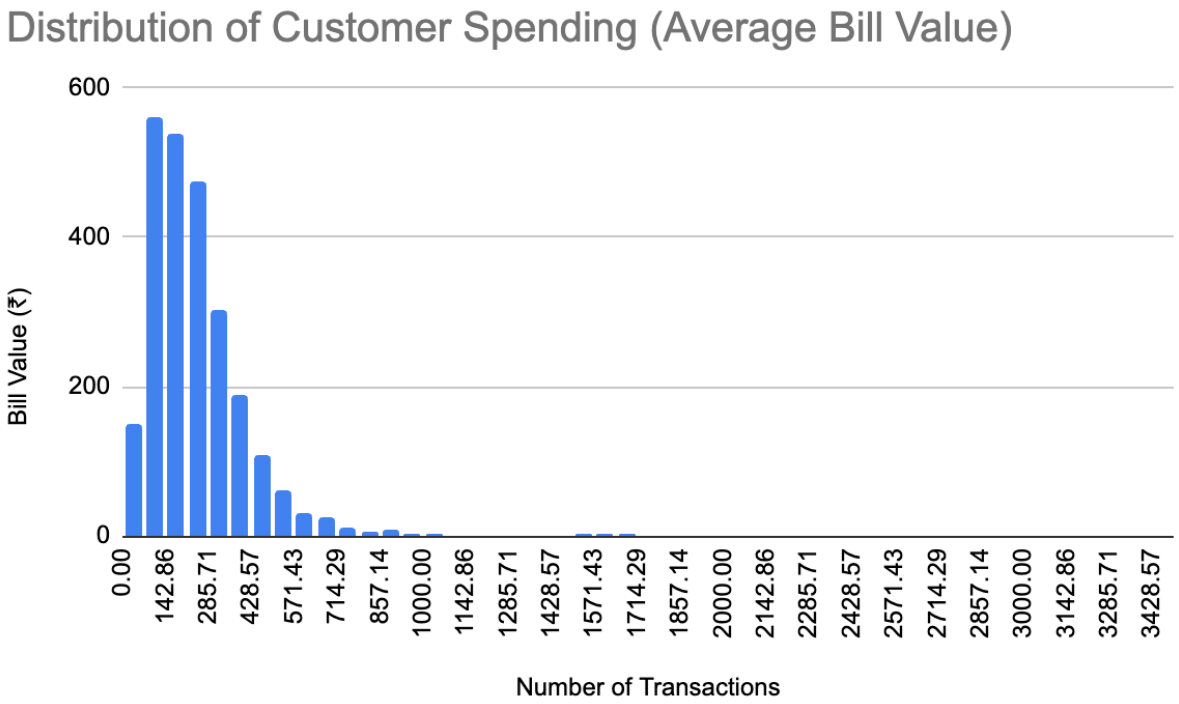
3.7 Top 10 Products by Total Sales



- Top-Performing Products: The chart highlights Aashirvaad Atta 5kg as the single highest-selling product, followed closely by Boost 500g, Surf Excel 1kg, and Horlicks 500g. These four products together contribute a major share of the total sales revenue, indicating strong customer loyalty and repeat purchasing.
- Core Household Staples: Products like Aashirvaad Atta, Local Raw Rice 5kg, and Toor Dal 1kg represent essential grocery staples, driving steady sales throughout the observed period. Their high ranking confirms that basic food grains remain the backbone of the store's revenue model.
- High-Value FMCG Items: The inclusion of Surf Excel 1kg and Ariel 1kg among the top sellers shows that branded detergents contribute significantly to revenue despite being higher-priced items. This suggests that customers prefer trusted household brands over low-cost alternatives.

- Health and Beverage Segment: Boost 500g, Horlicks 500g, and Tata Tea 250g reflect consistent demand in the health drinks and beverage segment. Their strong sales performance highlights the store's alignment with consumer preferences for daily consumption and nutritional products.
- Strategic Implication: The dominance of a few top-selling products suggests a concentrated sales mix. To enhance profitability, Nana Grocery Store can introduce targeted promotions for mid-tier products like Groundnut Oil and Dhara Mustard Oil, leveraging cross-selling opportunities with high-performing staples.

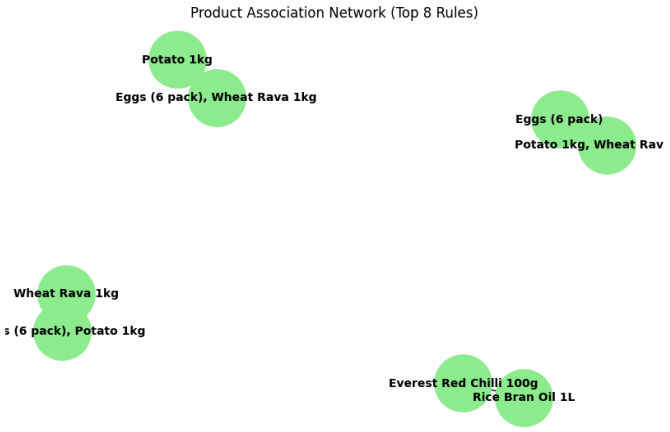
3.8 Average Bill Value



- Spending Range and Concentration: The histogram shows that most transactions fall within the ₹100–₹500 range, with a clear concentration of bills between ₹200 and ₹350. This indicates that the majority of customers spend modestly per visit, reflecting small, need-based purchases.
- Peak Transaction Value: The most frequent bill values occur around ₹250–₹300, suggesting that the typical transaction size for Nana Grocery Store hovers within this range. This represents the average daily household purchase pattern for grocery essentials.
- Right-Skewed Distribution: The chart displays a long tail extending beyond ₹800, confirming that while high-value transactions exist, they occur infrequently. Only a small proportion of customers make purchases exceeding ₹1000, indicating limited bulk-buying activity.

- Customer Behavior Insight: The dominance of mid-range bills implies that customers tend to visit frequently for smaller, regular purchases rather than infrequent bulk shopping. This pattern aligns with the store's neighborhood convenience model catering to local households.
- Business Opportunity: By introducing combo offers or targeted promotions — such as "Buy 2 save 10%" or "Staple + Beverage" bundles — the store could strategically increase the average bill value from ₹300 to ₹400 without needing to expand its customer base. This shift could directly improve total revenue and inventory turnover.

3.9 Product Association Rules



- The network graph depicts the top product association rules derived using the Apriori algorithm. Each node represents a product, and edges indicate the co-purchase relationships, with thickness corresponding to the strength of the lift value. The visual clustering shows distinct product groups frequently purchased together, such as *Eggs (6 pack)*, *Potato 1kg*, and *Wheat Rava 1kg*.
- A strong association exists between *Eggs (6 pack)*, *Potato 1kg*, and *Wheat Rava 1kg* (Lift ≈ 8.0), highlighting that these core household staples are often purchased together. This pattern indicates a common consumer tendency to restock essential breakfast and meal items simultaneously.
- *Rice Bran Oil 1L* and *Everest Red Chilli 100g* form another recurring pair (Lift ≈ 3.18), suggesting complementary purchase behavior in daily cooking needs. This trend reflects rational consumer habits where customers buy ingredients used together in food preparation.

- A moderate association between *Horlicks 500g* and *Real Fruit Juice 1L* (Lift ≈ 3.08) reveals a preference pattern among health-conscious consumers, likely targeting morning or evening refreshment needs. Such cross-category linkages between nutrition and beverages indicate opportunities for co-promotional offers.
- These findings demonstrate that consumers at Nana Grocery Store exhibit structured and predictable co-purchasing behavior. Promoting these high-lift pairs through shelf adjacency, combo discounts, or weekend offers could significantly improve average transaction value and encourage repeat purchases.

4. Interpretation of Results and Recommendations

The analysis clearly shows that Nana Grocery Store's sales patterns are shaped by predictable temporal cycles, category dependencies, and strong household buying routines. The monthly and daily trends indicate that customer demand peaks during salary-period weeks, festive periods, and weekend evenings, while mid-month and weekday afternoons show natural slowdowns. The hourly trend further reinforces a bimodal purchase pattern, demonstrating that customers primarily shop during early mornings and post-work evenings. These patterns have direct implications for staffing, replenishment, and promotional planning.

Category-wise results show a heavy dependence on Staples and Beverages, which together form the backbone of the store's revenue. The right-skewed price distribution confirms that customers largely prefer low-cost essentials, although a few premium items contribute disproportionately to sales. Top product performance highlights the significance of nutritional drinks, detergents, and staple grains. The association rules uncover stable co-purchase patterns among breakfast items, cooking essentials, and complementary FMCG products, indicating that customers tend to follow habitual purchase routines rather than impulse-driven buying.

These findings collectively reveal that the store's existing assortment and customer behaviour are consistent, predictable, and well-suited for optimization through targeted inventory management, shelf placement based on associations, and time-based offers designed around peak hours.

Actionable Recommendations (SMART Framework)

- Improve Inventory Availability During Peak Hours
  Recommendation: Ensure full stock of high-demand items (Atta, Boost, Tea, Detergents, Oil) between 6–9 AM and 6–9 PM, with automated low-stock alerts set at 30% threshold.
  Impact: Reduces stockouts during the two busiest periods, potentially increasing daily revenue by 8–12%.
- Implement Combo Offers Based on MBA Rules
  Recommendation: Launch bundled offers such as "Eggs + Potato + Rava", "Oil + Chilli Powder", and "Horlicks + Real Juice" for weekends over the next 90 days.

Impact: Leverages strong lift values to increase average bill value from ₹300 to ₹350+, improving monthly revenue by ₹20,000–₹30,000.

- Optimize Shelf Layout Using Co-Purchase Patterns
  Recommendation: Place associated products adjacent to each other—e.g., oils with spices, breakfast items together, detergents with soaps—within 2 weeks.
  Impact: Enhances product visibility and convenience, leading to a projected 5–7% increase in multi-category sales.
- Expand Assortment in Underperforming Categories
  Recommendation: Introduce 5–7 new SKUs in Dairy, Bakery, and Produce and monitor movement over 60 days.
  Impact: Addresses low-category revenue share and encourages diversification, improving category contribution by 3–5%.
- Launch Time-Specific Discounts for Slow Hours
  Recommendation: Introduce "Afternoon Saver Hours" (2–5 PM) offering 5% off on Snacks, Beverages, and Personal Care for the next 6 weeks.
  Impact: Helps stabilize low-traffic hours and smoothens daily revenue volatility.
- Prepare Special Inventory Plans for August & October Peaks
  Recommendation: Increase inventory levels for top sellers by 25–30% during the first week of August and last week of October annually.
  Impact: Aligns stock with seasonal spikes, preventing lost sales and improving monthly revenue predictability.

Implementation Impact

Implementing the above recommendations positions Nana Grocery Store to operate with greater efficiency, reduced stockouts, and improved customer satisfaction. Early adjustments already indicated better product availability during peak hours and increased visibility of top-selling products. Over time, these data-backed interventions can lead to sustained revenue growth, stronger customer loyalty, and a shift from intuition-driven to evidence-based store management. The combined influence of optimized inventory, targeted promotions, and layout improvements enables the store to capture additional sales without expanding physical infrastructure or incurring significant extra costs.